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EXPERIMENT 10

Aim: Identity and Access Management (IAM).

Theory:

Identity Access Management (IAM) is a framework of policies, technologies, and practices that ensure the right individuals (or entities) have the appropriate access to resources within an organization, while maintaining security and compliance. IAM is essential for managing digital identities, user authentication, and authorization across systems, networks, and applications.

Key components of IAM:

1. **Identity Management:** Creating, managing, and deleting user identities (e.g., employees, customers) within an organization's system.
2. **Authentication:** Verifying the identity of users or systems, typically using usernames, passwords, biometrics, or multi-factor authentication (MFA).
3. **Authorization:** Determining which resources a user is allowed to access, often controlled by roles, permissions, and policies.
4. **User Lifecycle Management:** Managing the entire lifecycle of a user's access, from onboarding to deactivation.
5. **Single Sign-On (SSO):** A service that allows users to access multiple applications with one set of login credentials.
6. **Audit and Reporting:** Monitoring access patterns and generating reports to ensure compliance with security policies and regulations.

In Google Cloud, **Identity and Access Management (IAM)** allows you to control who can access your cloud resources and what actions they can perform. IAM helps you securely manage cloud resources by assigning roles to users, groups, or service accounts, and specifying permissions for those roles.

Screenshots:

Step 1:



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



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You need additional access

You need additional access to the project: **CCLTY2025**

This could be because you have insufficient permissions to access the resource, or because a Principal Access Boundary Policy is blocking your access to the resource.

To request access, contact your project administrator and provide them a copy of the following information:

Troubleshooting info:

Principal: janhavinate17@gmail.com
Resource: cclty2025
Troubleshooting URL: console.cloud.google.com/iam-admin/troubleshooter?permissions=compute.instance

Missing permissions:

compute.instances.list

If your administrator is unable to help, then [contact support](#).

Step 2:

IAM

Permissions for project CCLTY2025

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

VIEW BY PRINCIPALS **VIEW BY ROLES**

GRANT ACCESS **REMOVE ACCESS**

Filter Enter property name or value

Type	Principal	Name	Role
☐	714692080618-compute@developer.gserviceaccount.com		Editor
☐	cclservice@cclty2025.iam.gserviceaccount.com		Browser
☐	janhavinate17@gmail.com	Janhavi Nate	Browser
☐	rohiniadhavphd@gmail.com	Rohini Bhosale	Owner

Conclusion:

Thus we successfully implemented Identity and Access Management (IAM) in Google Cloud.